

Applied Deep Learning

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Machine Learning Basics



Machine Learning Algorithms

“A computer program is said to learn from **experience** E with respect to some class of **tasks** T and **performance measure** P , if its performance at tasks in T , as measured by P , improves with experience E ”



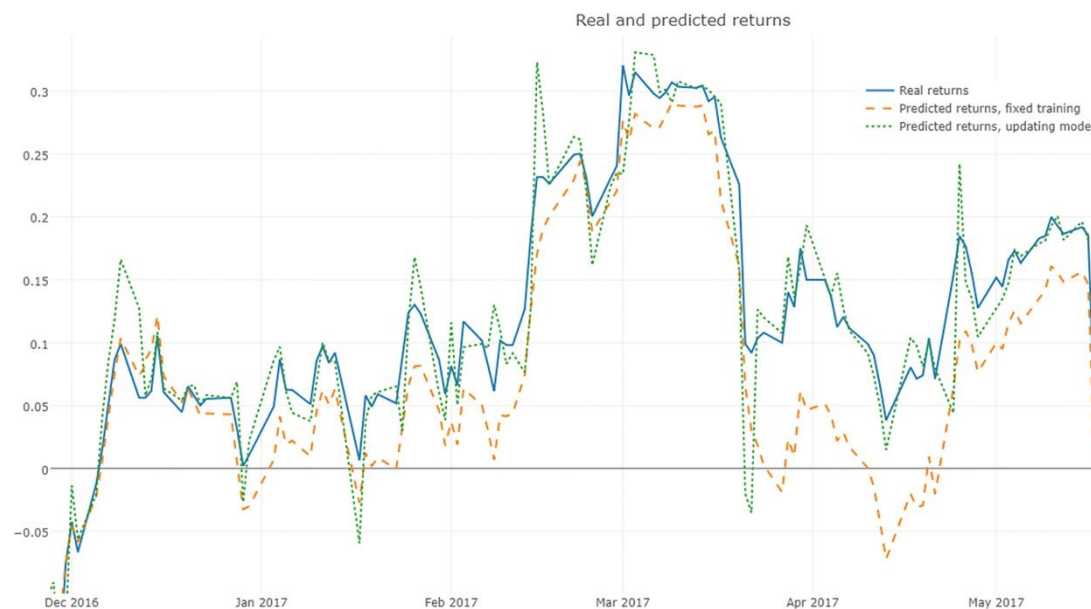
Tasks

image classification



CAT

stock regression

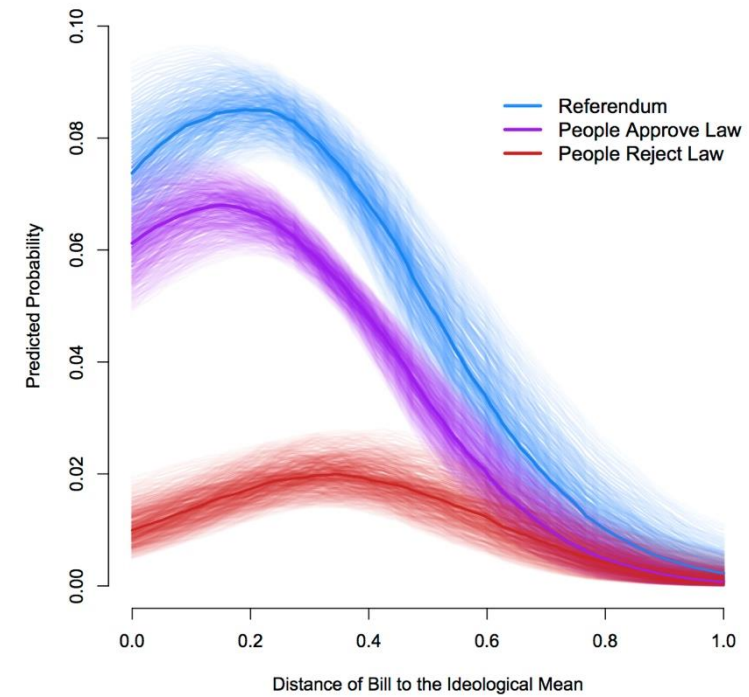


Performance Measures

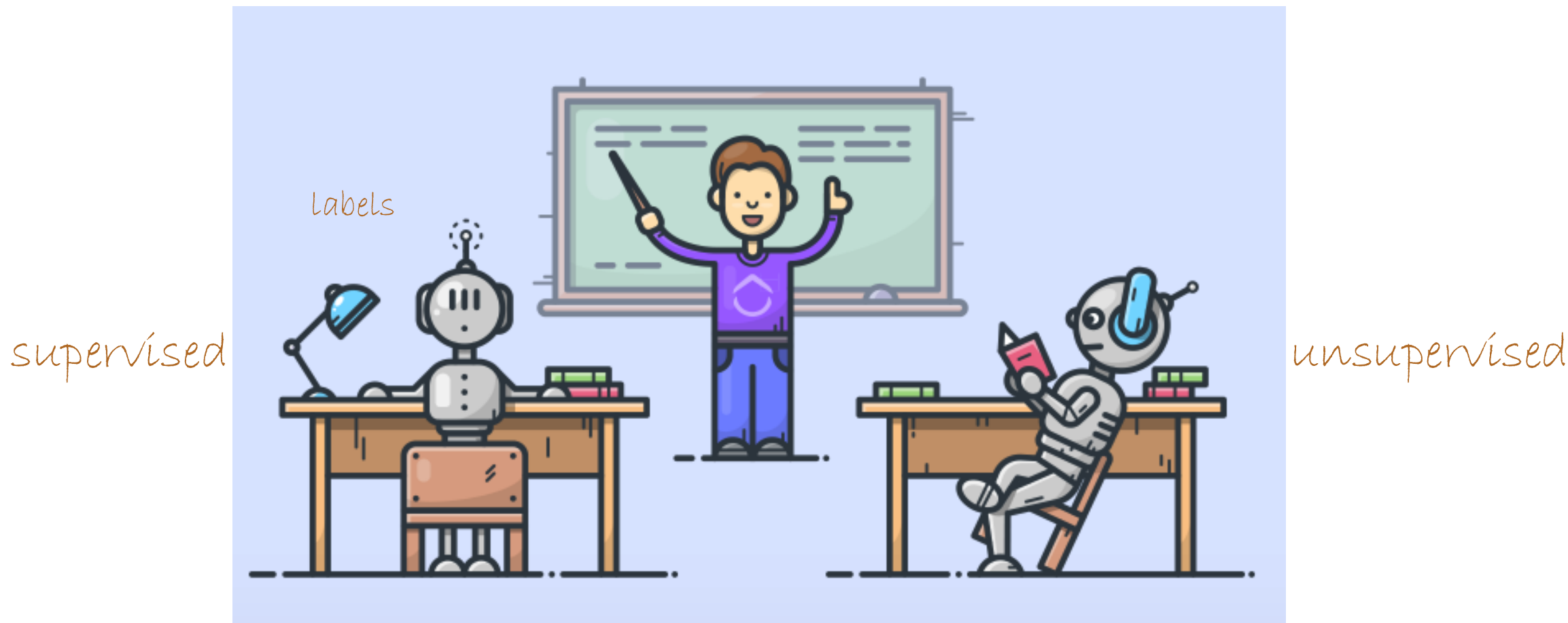
accuracy



distance



Tasks Classification



Duality of Problems

Unsupervised $\rightarrow$ Supervised

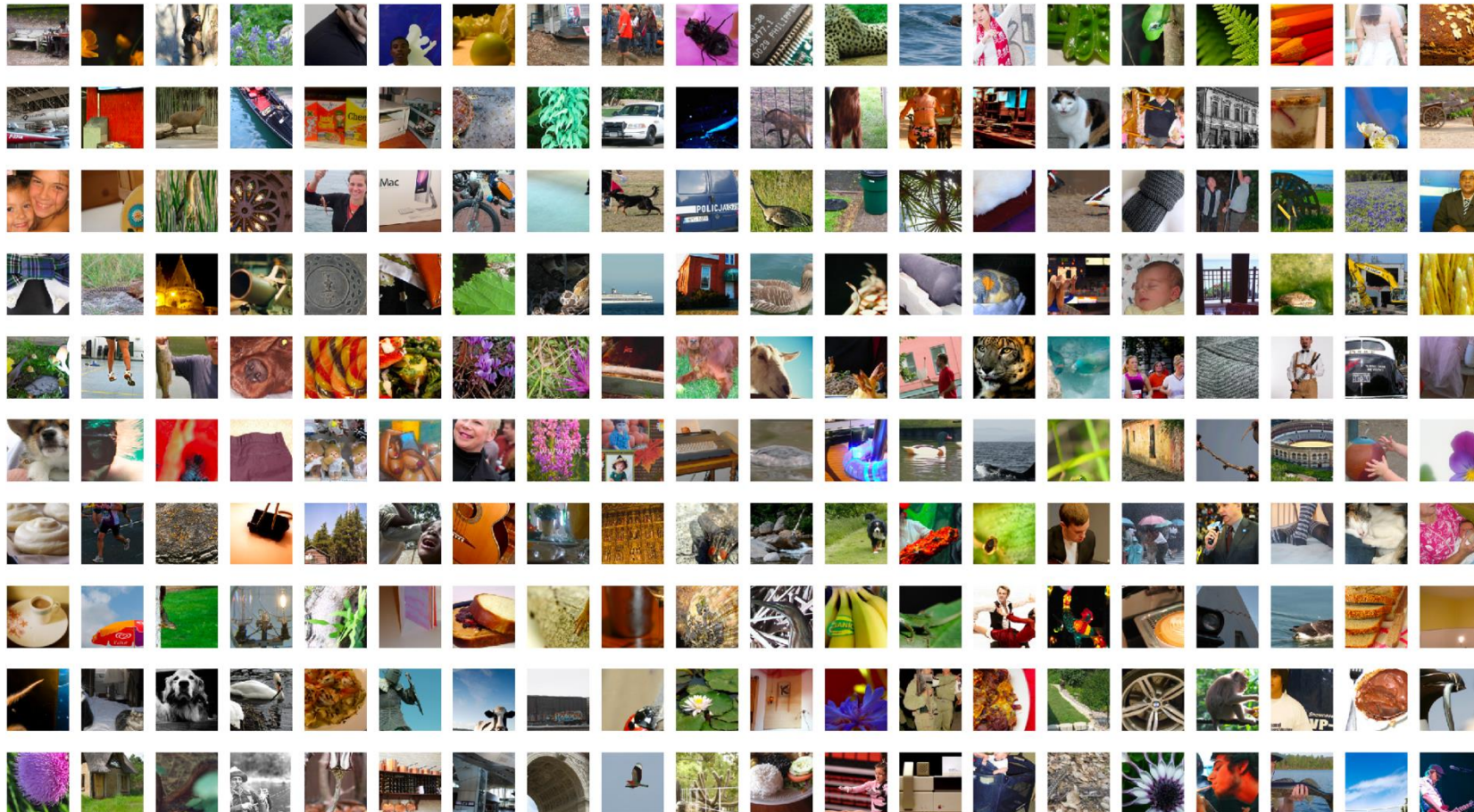
$$p(\mathbf{x}) = \prod_{j=1}^n p(\mathbf{x}_j \mid \mathbf{x}_1, \dots, \mathbf{x}_{j-1})$$

Supervised $\rightarrow$ Unsupervised

$$p(y \mid \mathbf{x}) = \frac{p(\mathbf{x}, y)}{\sum_{y'} p(\mathbf{x}, y')}$$



Experiences



Experiences

Training Set



80%

$$S = \{(x, y)\}$$

input

label

Test Set



20%

$$S' = \{(x', y')\}$$

i.i.d Assumption

train set



test set



Example: Linear Regression

Given a **train** set $\{(x, y)\}$, we consider

$$\hat{y} = w^T x + b$$

Diagram illustrating the linear regression equation $\hat{y} = w^T x + b$ with handwritten annotations:

- $\hat{y}$ is labeled "prediction" (with an arrow pointing to $\hat{y}$).
- $w^T x$ is labeled "parameters" (with an arrow pointing to $w^T x$).
- b is labeled "bias" (with an arrow pointing to b).

The **task** is to make $y \approx \hat{y}$

Example: Linear Regression

Get the best performance on a **test** set $\{(x', y')\}$

$$\text{MSE}_{\text{test}} = \frac{1}{2} |w^T x' - y'|_2^2$$

$\hat{y}'$

We **solve** for the train set $\min_w \text{MSE}_{\text{train}}$



Example: Linear Regression

The **necessary** condition reads

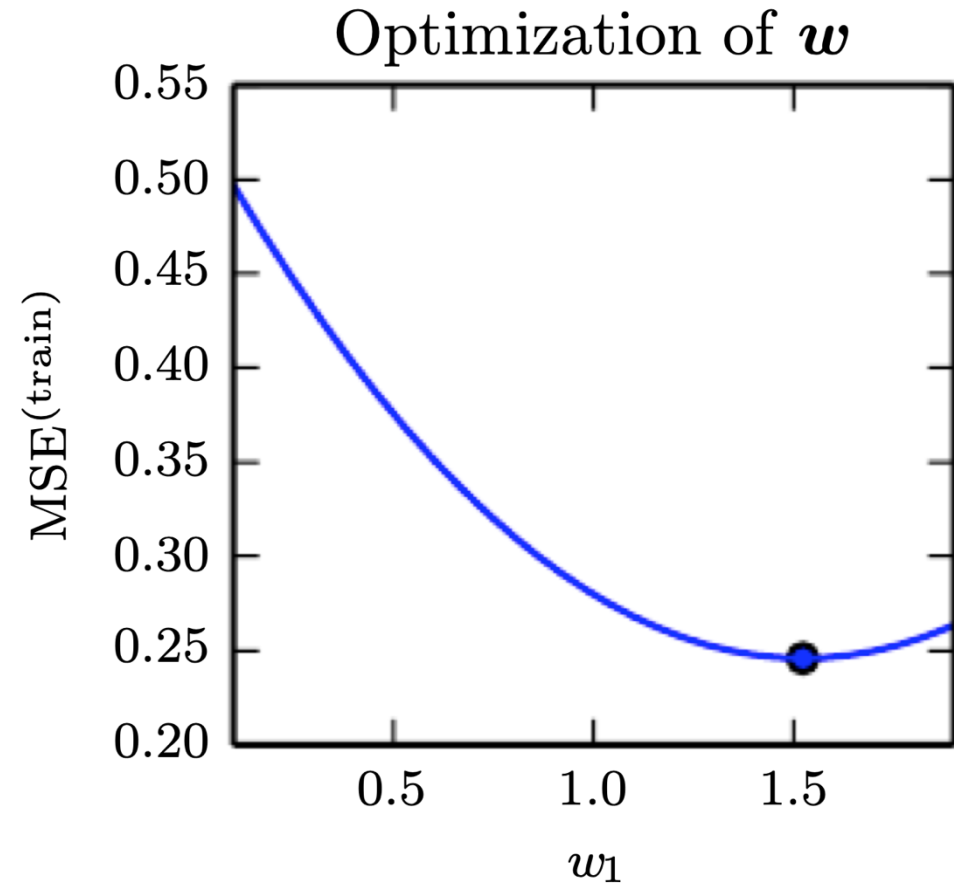
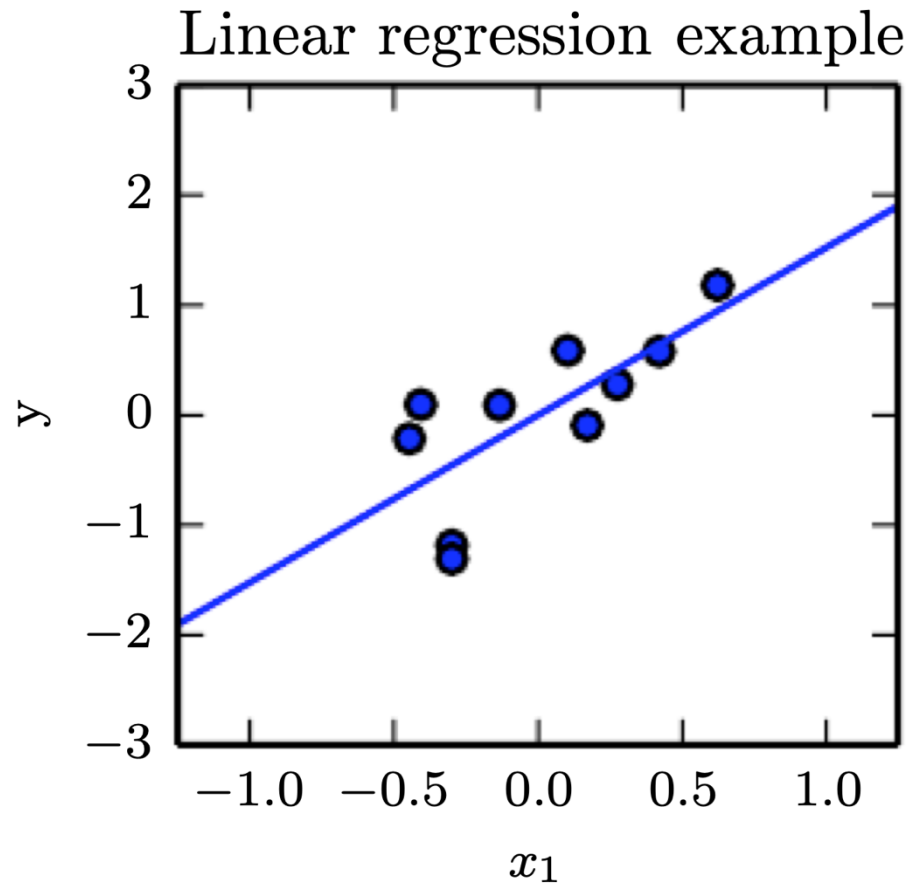
$$\nabla_w \text{MSE}_{\text{train}} = 0$$

Thus

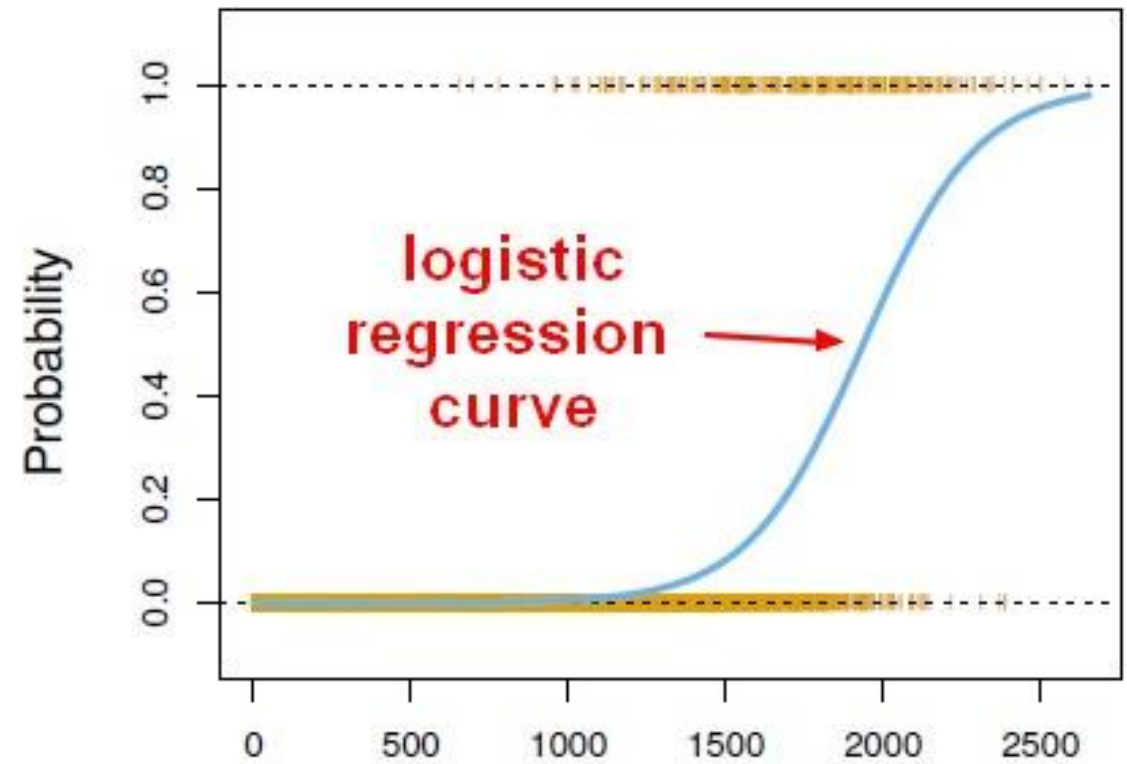
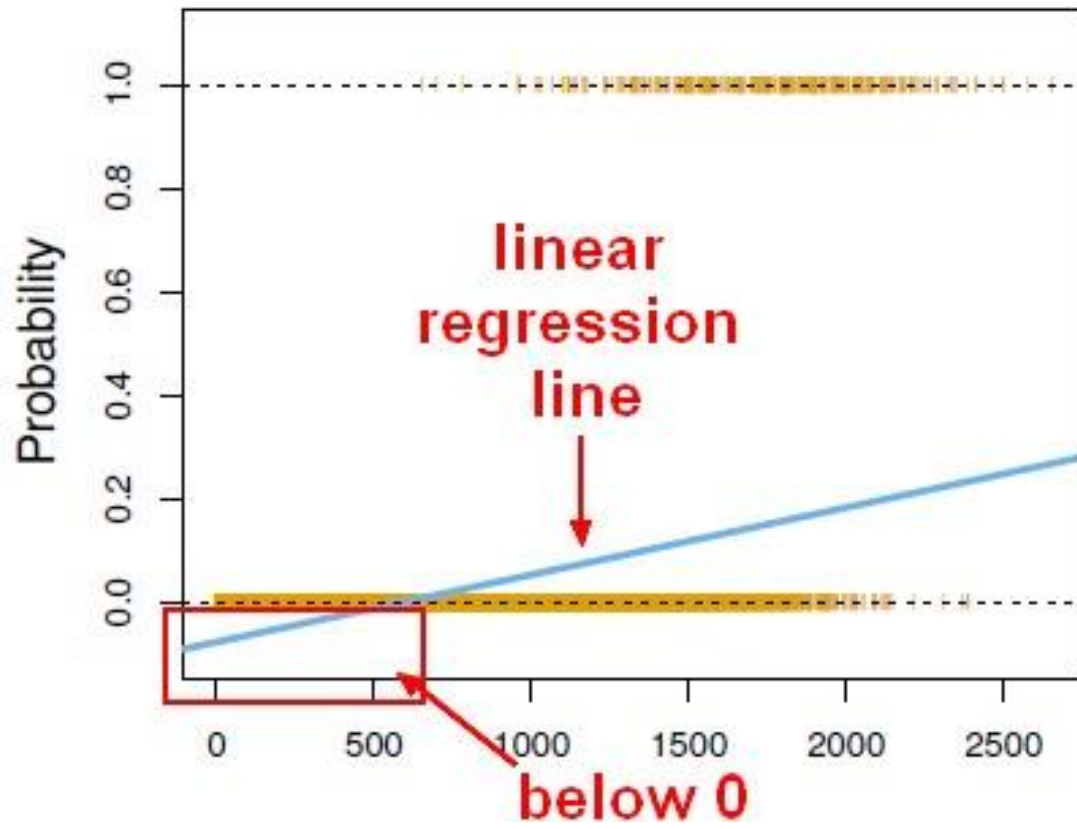
$$XX^T w - X^T y = 0 \implies w = X^+ y$$



Example: Linear Regression

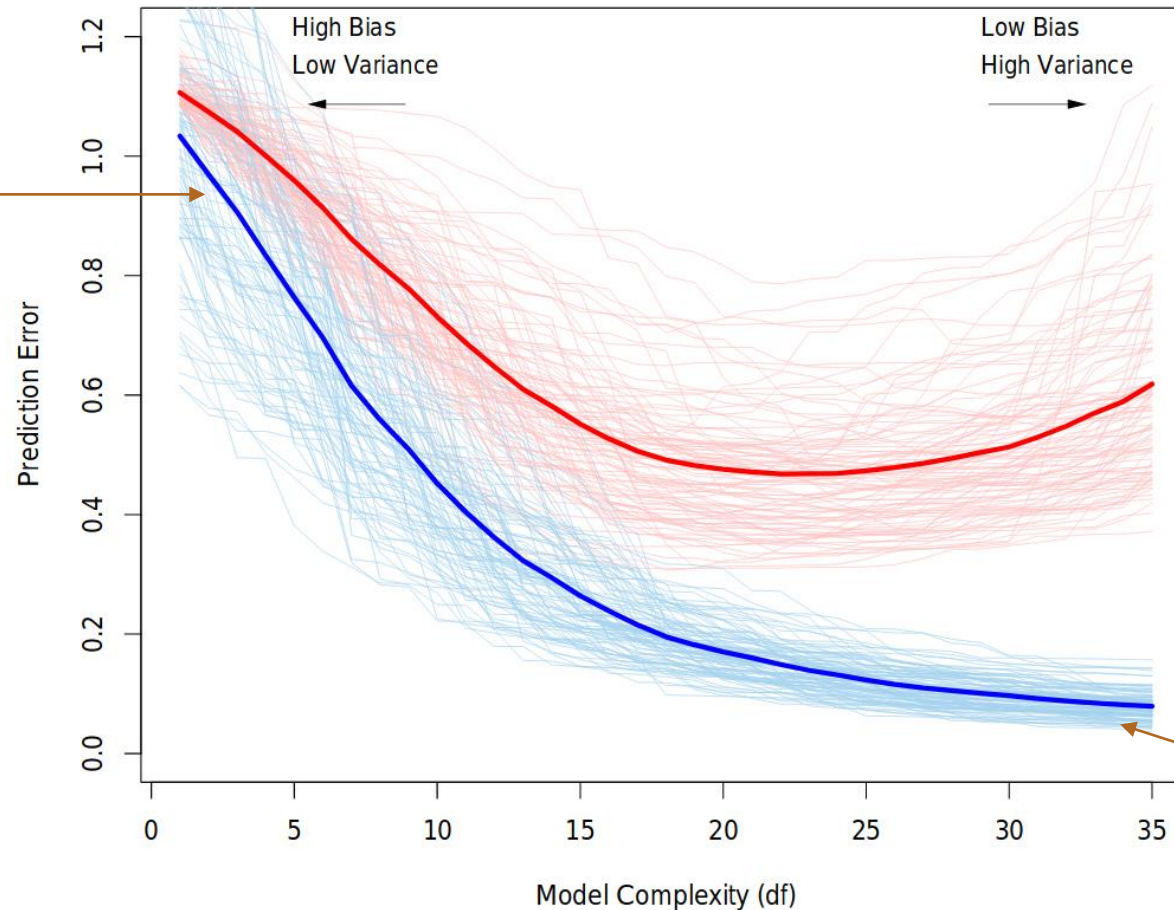


Linear vs. Logistic Regression



Capacity, Overfitting, and Underfitting

underfitting



generalization gap

overfitting



ML No Free Lunch Theorem

“There is no **single** ML algorithm that is universally the best-performing algorithm for **all** problems”



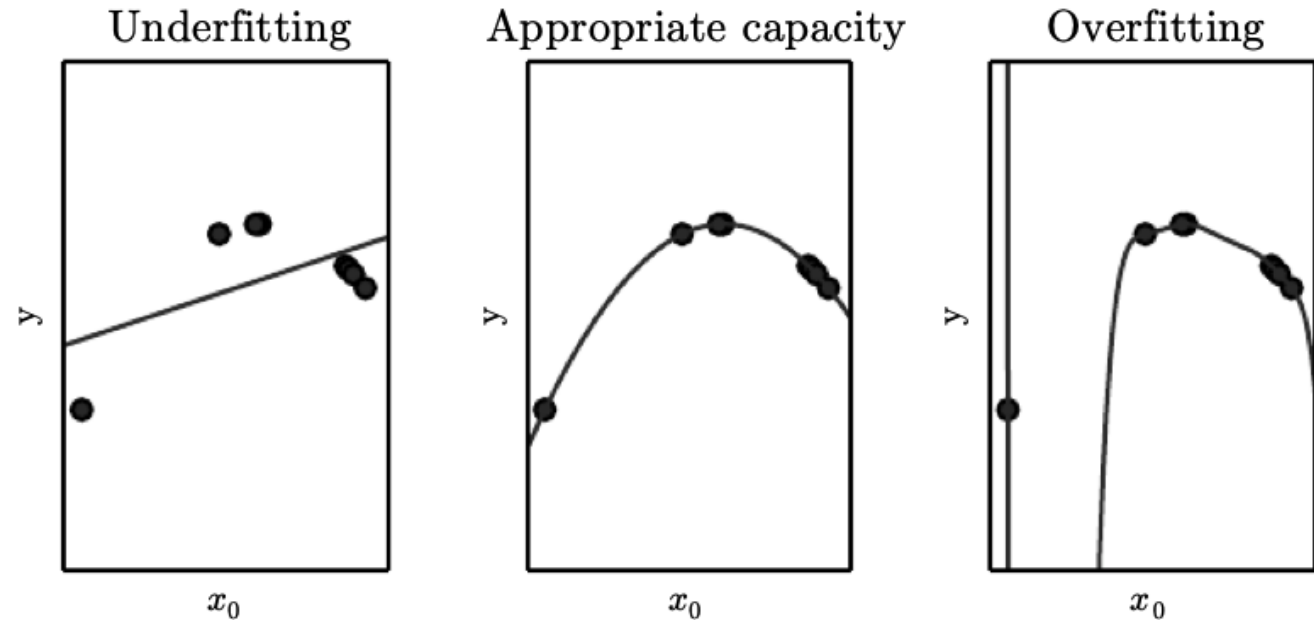
Hypothesis Space

Linear

$$\hat{y} = b + wx$$

Quadratic

$$\hat{y} = b + w_1x + w_2x^2$$



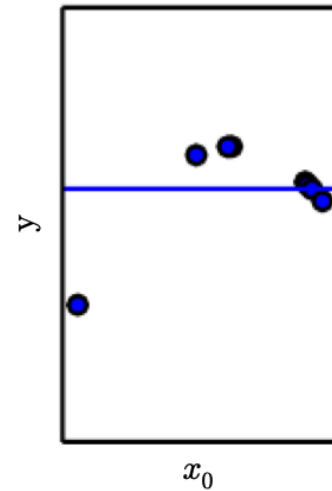
Regularization

Structured hypothesis space

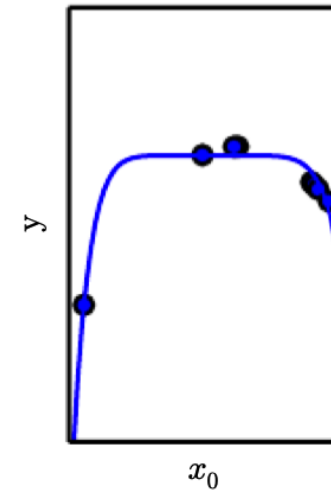
$$\min_w \text{MSE}_{\text{train}} + \lambda w^T w$$

↑
weight decay

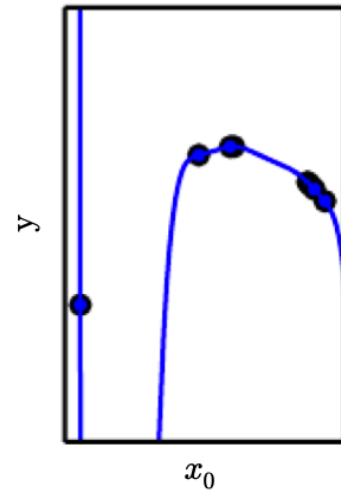
Underfitting
(Excessive λ)



Appropriate weight decay
(Medium λ)

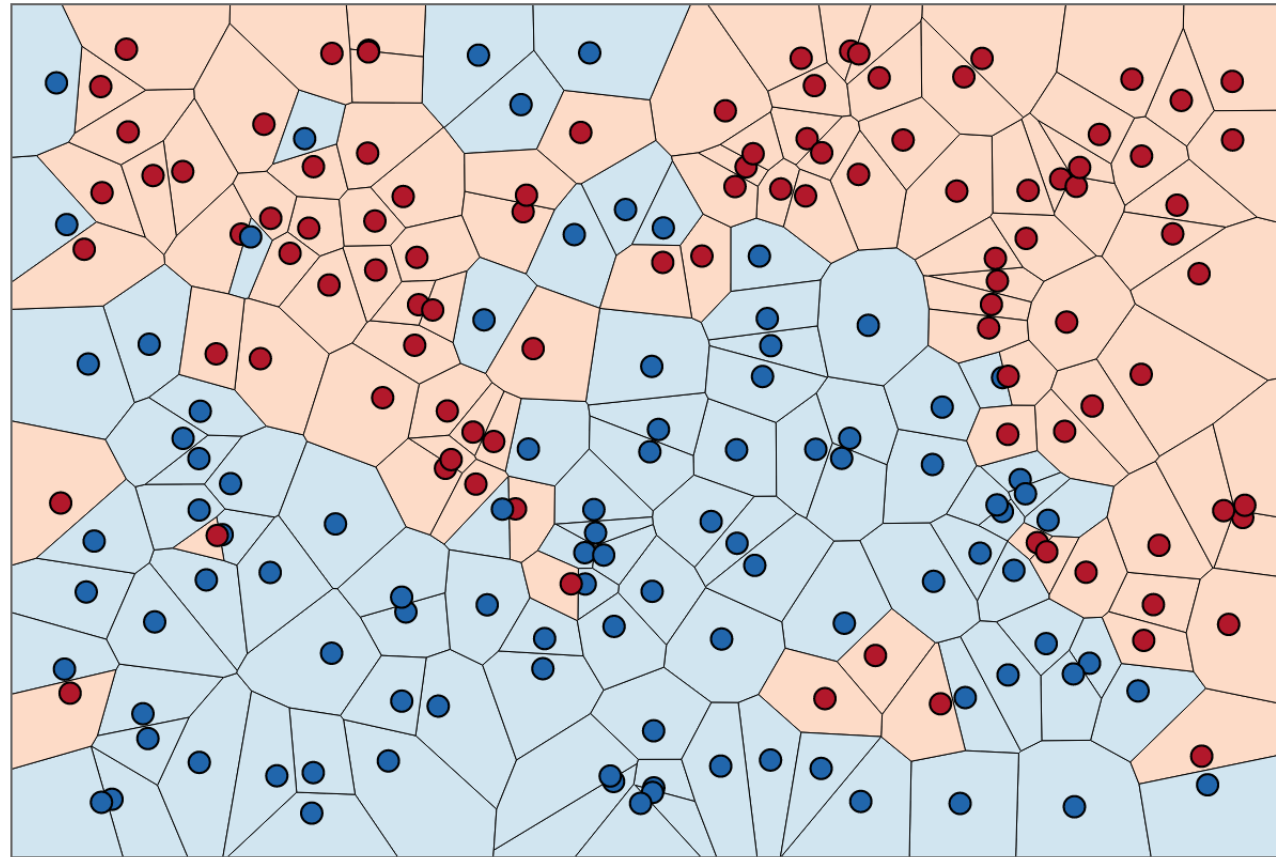


Overfitting
($\lambda \rightarrow 0$)



Example: Nonparametric Models

nearest
neighbors



Hyperparameters

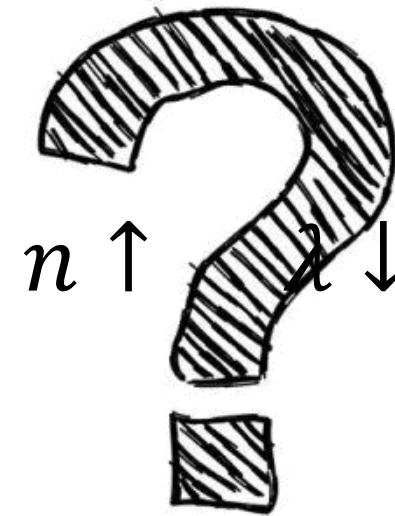
Degree of polynomial

$$\hat{y} = b + w_1x + \dots + w_nx^n$$

Regularization scale

$$\min_w \text{MSE}_{\text{train}} + \lambda w^T w$$

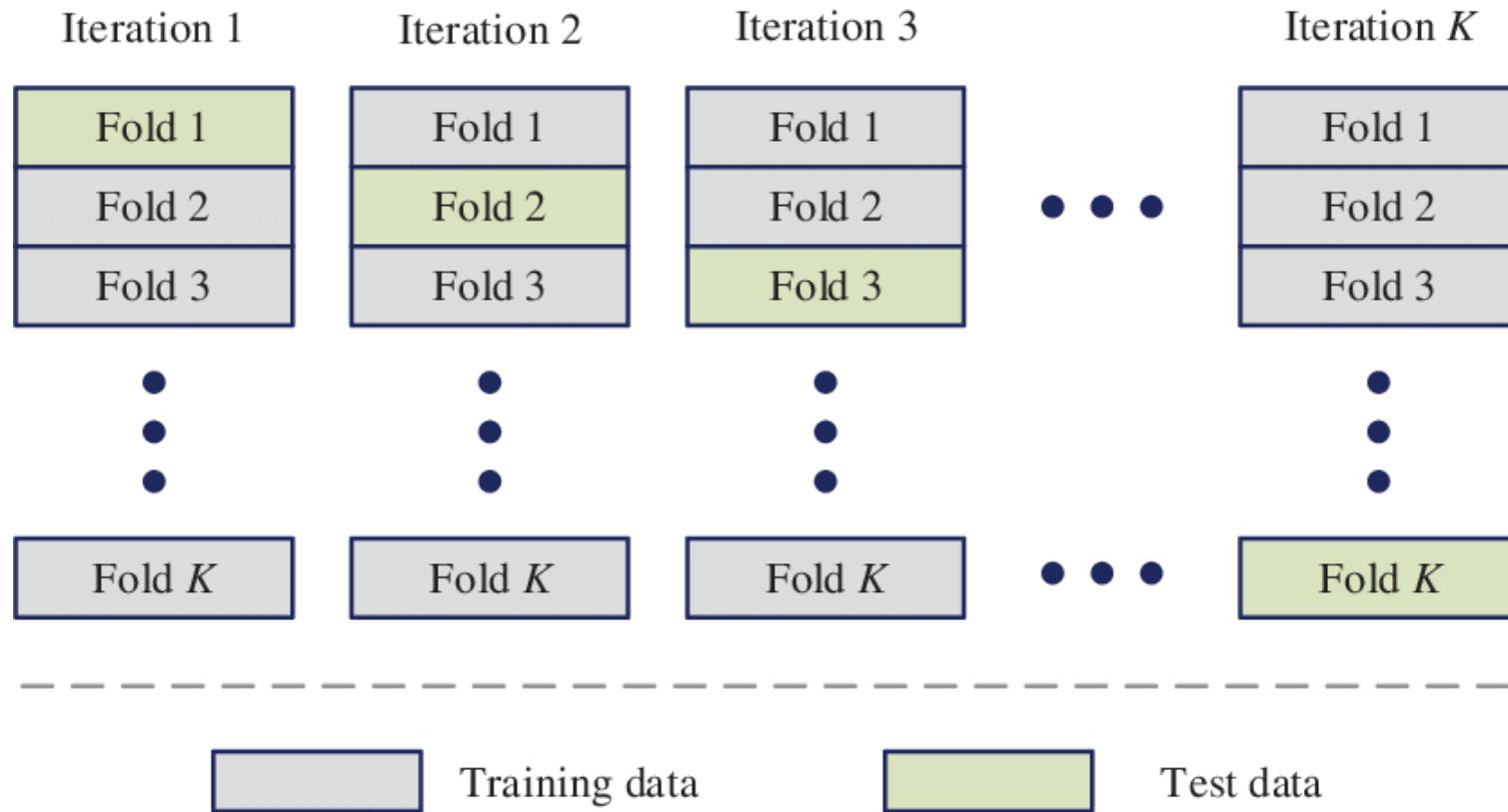
Use hyperparameter **weights**?



Validation Set



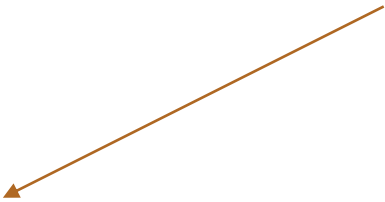
Cross-Validation



Estimators: Bias and Variance

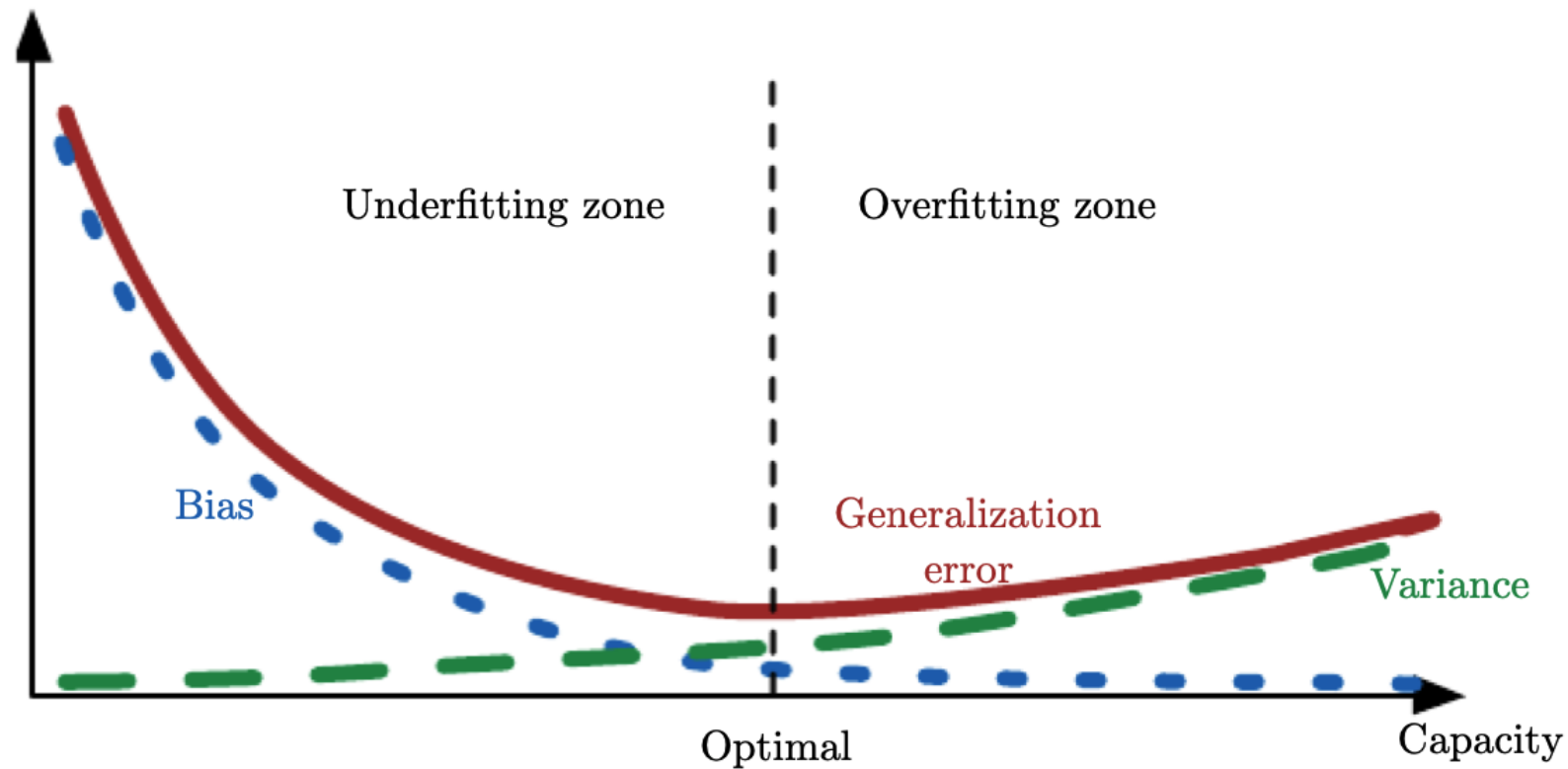
We obtain that

$$\text{MSE} = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$


$$\mathbb{E}(\hat{\theta}) - \theta$$


$$\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$$

Bias and Variance Tradeoff



Maximum Likelihood Estimation

How to design **good** estimators?

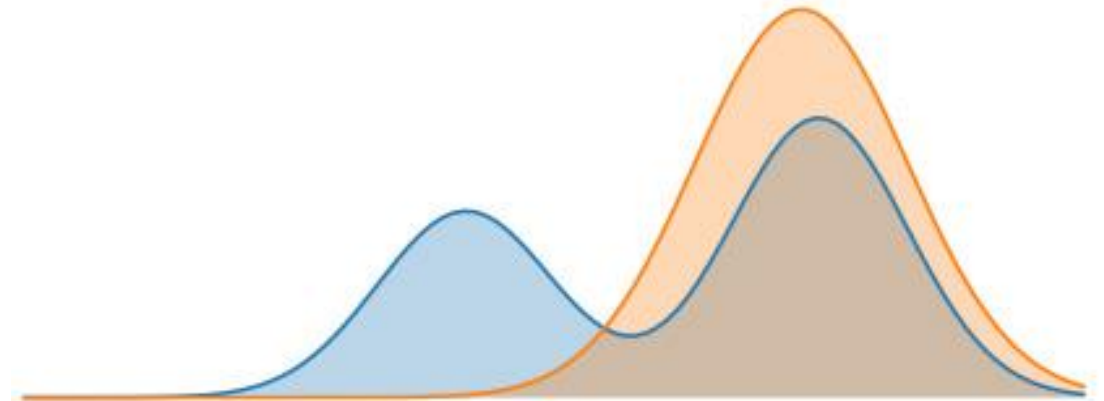
$$\theta_{\text{ML}} = \arg \max_{\theta} p_{\text{model}}(\mathbb{X}; \theta)$$



$$\theta_{\text{ML}} = \arg \max_{\theta} \mathbb{E}_{x \sim \hat{p}_{\text{data}}} \log p_{\text{model}}(x; \theta)$$



(negative) log-likelihood



KL Divergence

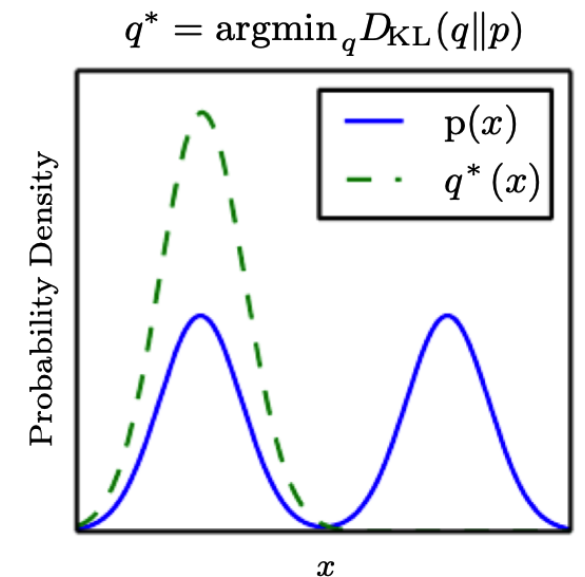
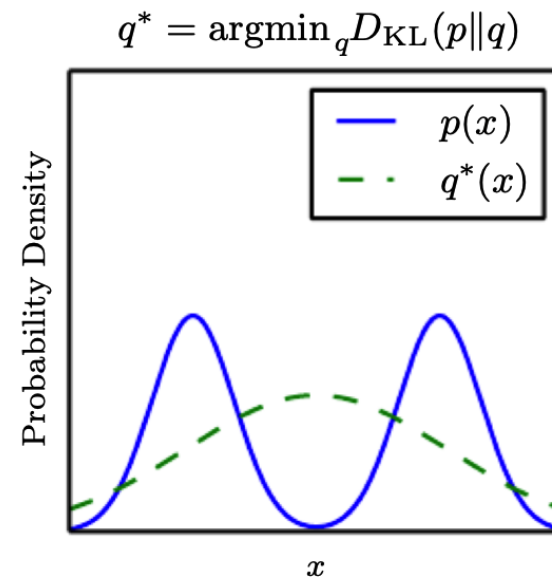


Distance between Distributions

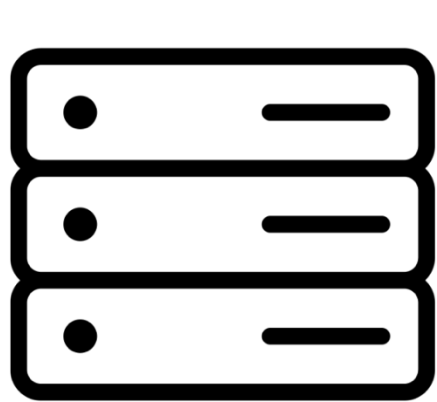
Kullback—Leibler divergence

$$D_{KL}(P \parallel Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

↑
asymmetric



Build an ML Algorithm



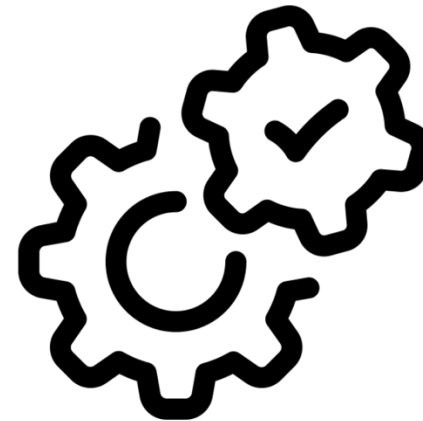
images

+



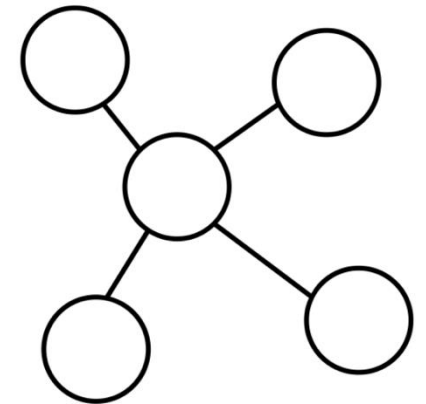
MSE

+



SGD

+



linear regression



The Curse of Dimensionality

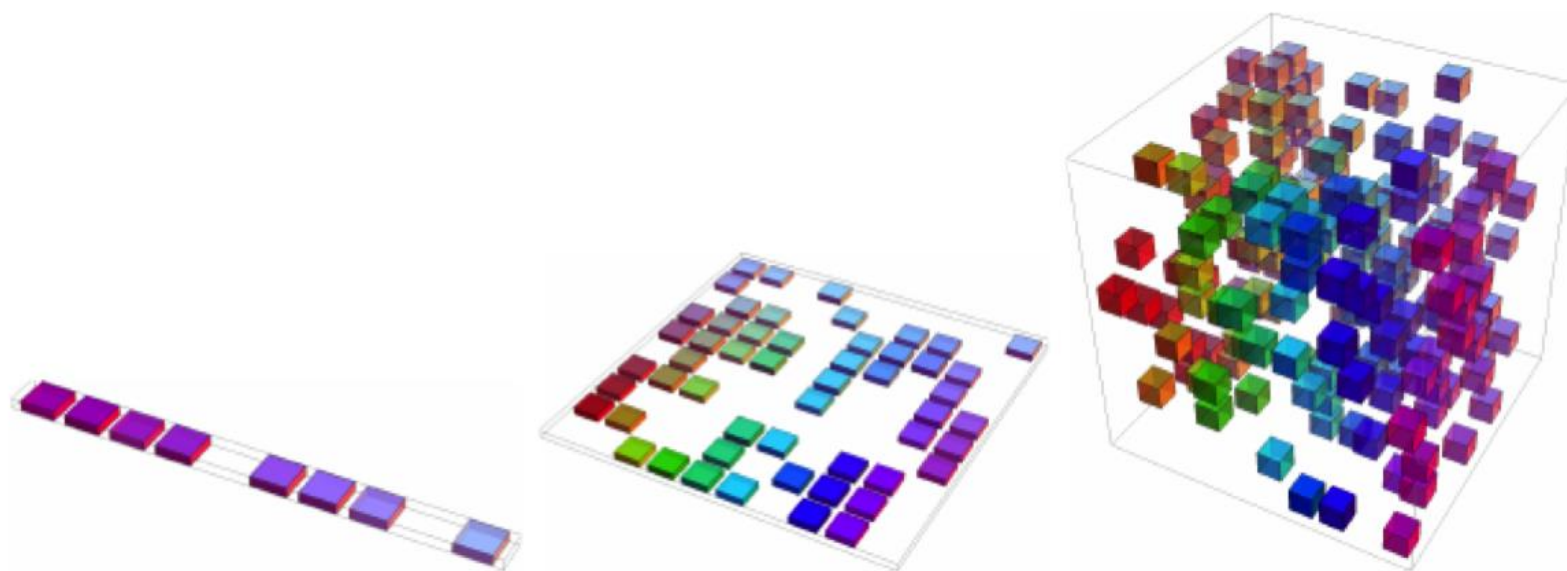


ינון מגל
@YinonMagal

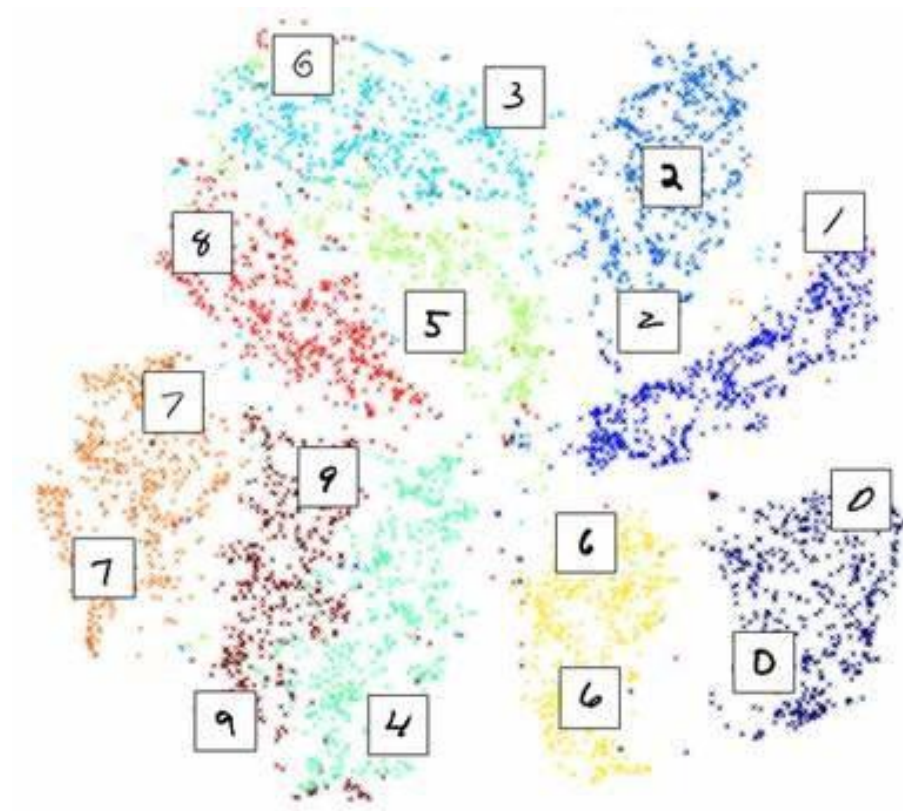
נס גאוגרפי:
לטענת מארגני ההפגנות השתתפו בהן
הערב 210,000 איש
אם שמרו על ההנחיות, ועל מרחק של שני
מטר - זה סהכ 420 קילומטר
מזל שלא גלשו ללבנון בטעות



The Curse of Dimensionality



Manifold Learning



Code Demo

